EE 604T – Pattern Recognition and Machine Learning
Project Report: Classification, Regression and Clustering on a Synthetic Loan Dataset

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***Abstract*—In this project, we explore and compare various machine learning models for three tasks: classification, regression, and clustering. The dataset contains features describing the financial situation and personal attributes of individuals. In classification, the goal is to predict whether a loan is approved. For regression, models estimate an individual's risk score. Clustering identifies groups in the data, using the Elbow Method and Silhouette Analysis to find the optimal number of clusters.**

***Index Terms*—classification, regression, clustering, PCA, loan approval, risk score**

I. Introduction

We use a synthetic dataset of 20,000 records of personal and financial data. The dataset has thirty-six columns, grouped as follows:

- **Demographic and employment:** Age, Marital Status, Number of Dependents, Education Level, Employment Status, Experience, Job Tenure.
- **Financial and credit:** Annual/Monthly Income, Total Assets, Total Liabilities, Net Worth, Home Ownership Status, Savings and Checking Balances, Debt-to-Income Ratio, Total Debt-to-Income Ratio, Monthly Debt Payments, Utility Bills Payment History.
- **Credit history and loans:** Credit Score, Credit Card Utilization Rate, Number of Open Credit Lines, Number of Credit Inquiries, Bankruptcy History, Length of Credit History, Previous Loan Defaults, Payment History.
- **Loan attributes:** Loan Amount, Loan Duration, Loan Purpose, Base Interest Rate, Interest Rate, Monthly Loan Payment, Application Date.
- **Targets:** Loan Approved (classification) and Risk Score (regression).

II. Methodology

A. Analysis and Preprocessing

Duplicate and missing values. No duplicate rows or rows with missing values were found.

Categorical columns. Employment Status, Education Level, Marital Status, Home Ownership Status and Loan Purpose were converted to integer labels and, separately, to one-hot encodings, producing two dataset variants (33 and 49 feature columns respectively).

Split. A 60-20-20 split gave 12,000 training examples and 4,000 each for validation and test.

Outliers. The boxplots (Fig. 1) show many outliers, mainly in net worth, total assets, total liabilities and income. As these are valid data points, they were retained.

Scaling. Min-max, Z-score and Robust scaling were each applied separately.

Dimensionality reduction. `ApplicationDate` was dropped since it does not affect the targets. PCA reduced the integer-labeled data from 33 to 10, 15, 20 and 25 dimensions, and the one-hot data from 49 to 15, 20, 25, 30 and 35 dimensions (Fig. 5).

B. Exploratory Visualizations

The correlation heatmap (Fig. 2) and scatter plots (Fig. 3) show strong linear relationships between related features, such as net worth and total assets, and monthly and annual income. Monthly and annual income also correlate most strongly with the targets (Fig. 4).

III. Model Implementation and Comparison

A. Classification

The target `LoanApproved` was predicted using k-Nearest Neighbors, Logistic Regression, Support Vector Machines and Artificial Neural Networks.

k-NN (Mahalanobis variant). Using `KNeighborsClassifier` with $k = 3, 5, 7, 11$, the best accuracy was 82.50% and precision 81.84% (at $k = 11$), best recall 44.35% and best F1-score 52.65%. For $k = 5$ the test metrics were accuracy 81.93%, precision 70.40%, recall 42.05%, F1 52.65%; for $k = 11$ they were 82.50%, 81.84%,

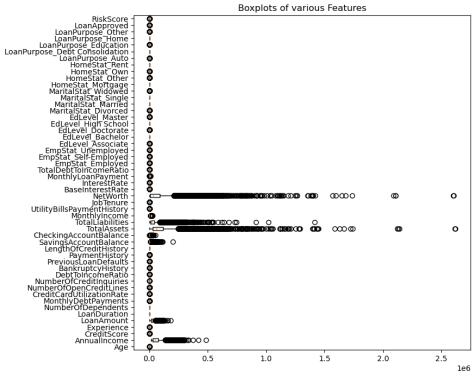


Fig. 1. Boxplots of various features.

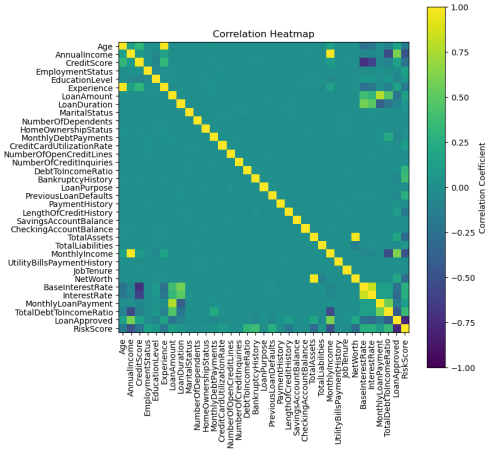


Fig. 2. Correlation heatmap between features.

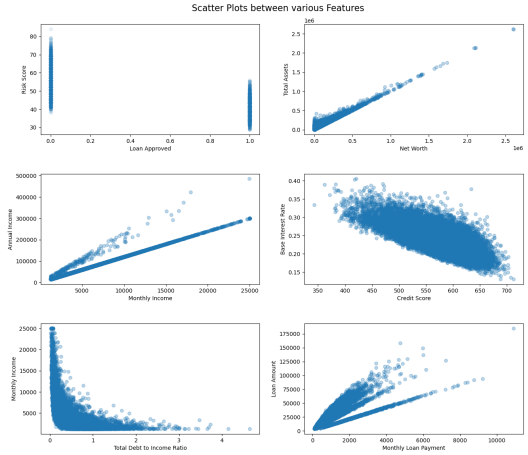


Fig. 3. Scatter plots between various features.

34.41% and 48.45%. The best dataset used one-hot encoding with Z-score normalization.

Logistic Regression. With `max_iter=1000`, the best accuracy (96.70%) and F1 (93.09%) were obtained on Z-score data with one-hot encoding. Best precision (93.98%) came from min-max scaled one-hot data and best recall

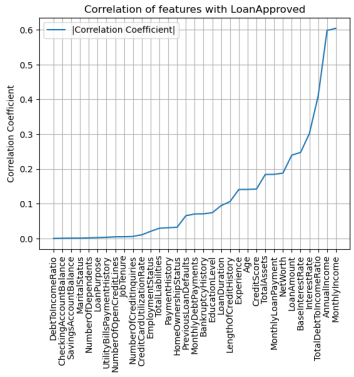


Fig. 4. Correlation of features with `LoanApproved`. Monthly and annual income correlate most strongly with the target.

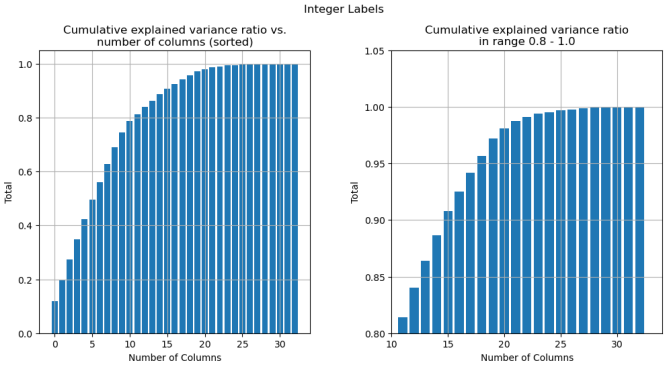


Fig. 5. PCA explained variance ratios for the integer-labeled dataset.

TABLE I
ANN TEST PERFORMANCE BY ARCHITECTURE (%)

Architecture	Acc.	Prec.	Rec.	F1
Single (128)	96.57	92.44	93.31	92.87
Double (8, 64)	96.75	92.93	93.51	93.22
Triple (8, 32, 64)	96.77	93.21	93.31	93.26

(92.99%) from Robust-scaled one-hot data. Test metrics: accuracy 96.70%, precision 93.19%, recall 92.99%, F1 93.09%.

SVM. `SVC` with $C \in \{0.1, 1, 10\}$ and `gamma='scale'` was tested; $C = 1$ was best for both kernels, on one-hot Z-score data.

- RBF kernel: accuracy 96.20%, precision 93.23%, recall 90.69%, F1 91.94%.
- Polynomial kernel: accuracy 94.20%, precision 94.80%, recall 80.13%, F1 86.85%.

Artificial Neural Network. A `torch.nn` network with BCE loss and the Adam optimizer was used, varying hidden layers from 1 to 3 and neurons from 8 to 128. Results are in Table I. The double-layer first-layer weights ranked Employment Status, Interest Rate and Number of Open Credit Lines highest, and Monthly Loan Payment, Base Interest Rate and Total Assets lowest.

Overall. One-hot encoded, Z-score normalized data per-

TABLE II
PERFORMANCE METRICS OF BEST CLASSIFICATION MODELS (%)

Model	Acc.	Prec.	Rec.	F1
Logistic Regression	96.70	93.19	92.99	93.09
ANN (triple layer)	96.77	93.21	93.31	93.26

TABLE III
REGRESSION TEST RMSE

Model	RMSE
Linear Regression	0.066
Polynomial (degree 2)	0.066
Random Forest	0.051
XGBoost	**0.046**
ANN (Z-score, double layer)	0.304

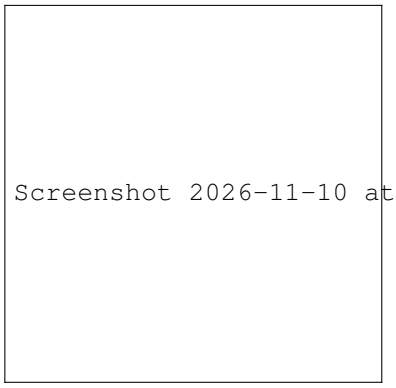


Fig. 6. Actual vs. predicted risk score for XGBoost on one-hot, min-max data.

formed best for almost all models. Logistic Regression and the triple-layer ANN were the best classifiers (Table II).

B. Regression

The target `RiskScore` was predicted with Linear, Polynomial, Random Forest, XGBoost and ANN regressors. Test RMSE values are summarized in Table III.

Linear and Polynomial Regression. Both achieved a test RMSE of 0.066 on one-hot, min-max normalized data (best polynomial degree: 2).

Random Forest. For `n_estimators`$=100, 200, 300, 500$, all gave similar validation MSE of 0.0042 and R^2 of 0.7950. Test RMSE was 0.051.

XGBoost. `XGBRegressor` with `reg:squarederror`, `max_depth=6` and `min_child_weight=10` gave the lowest test RMSE of 0.046 (Fig. 6).

ANN. On one-hot Z-score data, test RMSE was 0.325, 0.304 and 0.313 for single, double and triple layer architectures. The target is Z-score scaled here, so these values are not directly comparable to the min-max results.

C. Clustering

Three models were used: k-Medoid, DBSCAN and Agglomerative Hierarchical Clustering. For visualization, the two leading principal components were used.

TABLE IV
K-MEDOIDS RESULTS ($k = 2$), SELECTED DATASETS

Dataset	Silhouette	Inertia
One-hot, min-max (full)	0.0209	91323.67
One-hot, min-max, PCA-15	0.0712	48832.19
One-hot, min-max, PCA-35	0.0201	77578.42
Integer, min-max (full)	0.0242	70411.07
Integer, min-max, PCA-10	0.0688	36153.90
Integer, min-max, PCA-25	0.0267	62399.74

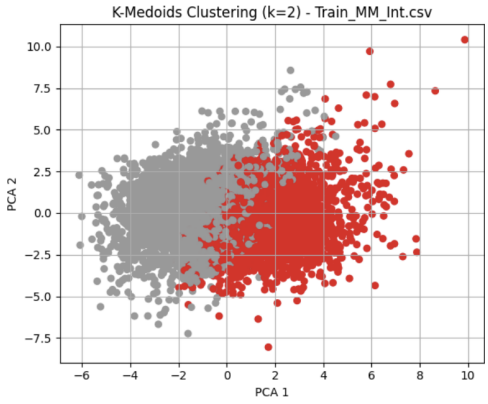


Fig. 7. k-Medoids clustering ($k = 2$) visualized on the first two principal components.

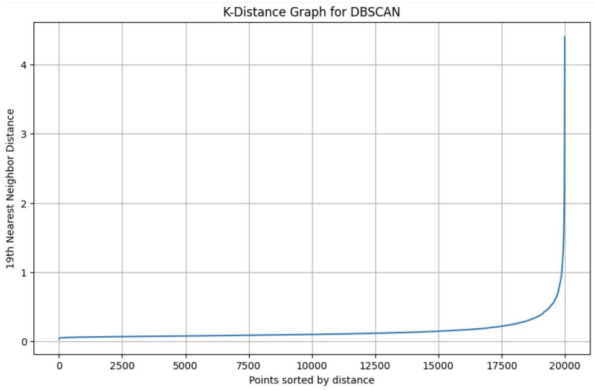


Fig. 8. k-distance graph and DBSCAN clustering result.

k-Medoid. The elbow method and silhouette scores were used to select k; the elbow curve was almost linear after $k = 2$, so $k = 2$ was used. Table IV lists results for selected datasets. The highest silhouette score (0.0712) was obtained on the one-hot, min-max PCA-15 dataset, followed by integer-labeled PCA-10 (0.0688). The resulting clusters are shown in Fig. 7.

DBSCAN. The k-distance graph (Fig. 8) was used to select `eps`; `eps` $= 0.5$ and `min_samples` $= 5$ were chosen. Because there is no low-density separation in the data, almost every point was grouped into a single cluster, a known caveat of DBSCAN.

TABLE V
PURITY SCORES FOR DIFFERENT LINKAGE METHODS

Linkage	Purity
Ward	0.85
Average	0.78
Complete	0.81
Single	0.65

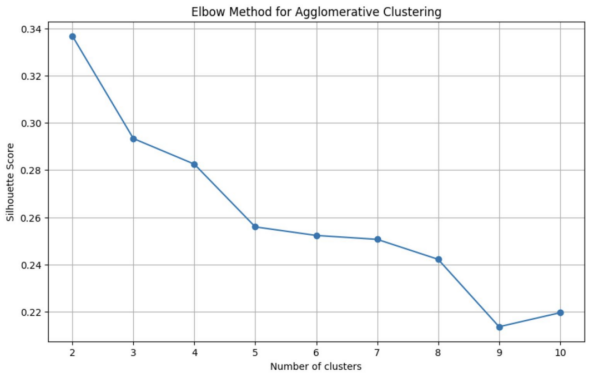


Fig. 9. Silhouette/elbow graph for agglomerative clustering.

Agglomerative Hierarchical Clustering. Four linkage methods were compared using purity score (Table V); Ward linkage performed best. The silhouette-based elbow plot (Fig. 9) peaks at two clusters.

IV. FINAL RESULTS AND CONCLUSIONS

Classification. One-hot encoding with Z-score normalization gave the best results. Logistic Regression and the triple-layer ANN performed best, reaching about 96.7% accuracy and 93% F1-score (Table II).

Regression. One-hot encoding with min-max normalization gave the best results. XGBoost was the most suitable model, with a test RMSE of 0.046.

Clustering. DBSCAN failed to identify clusters due to the lack of a low-density separating region. Agglomerative Clustering and k-Medoid Clustering were better suited to this dataset.